

CYCLIC EQUALITY-FEEDBACK DYNAMICS AT DYADIC LENGTHS

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ABSTRACT. Fix $q \geq 3$ and a dyadic length $n \geq 4$. We iterate on q -ary cyclic words the local rule which records whether adjacent letters are equal. The first step is nonlinear and q -ary, whereas the tail is the affine complement of periodic Rule 102. After assigning zero contribution credit to the additive cellular automaton, repeated-root code weights, and the cycle chromatic polynomial, we determine the remaining finite atlas. It contains a sharp height $n + 1$, all depth shells, the complete image staircase, and every target fibre as a q -weighted affine-code enumerator. Two slices of that inverse atlas are evaluated completely: time two is classified by the smaller weight in a complementary solution pair, and time $n/2 + 1$ by a half-word weight. Their parameter-class multiplicities are explicit, including possible collisions of numerical fibre values. An exact audit checks all words and targets in six boxes and the two special spectra through length 16.

1. OBJECT, PRIOR BOUNDARY, AND THEOREM

Let $\mathcal{A}_q = \{0, \dots, q - 1\}$ and index coordinates by $\mathbb{Z}/n\mathbb{Z}$. Define

$$(1) \quad T_q(w)_i = \mathbf{1}\{w_i = w_{i+1}\}, \quad w \in \mathcal{A}_q^n.$$

Thus $T_q(w)$ is binary after one step. Let S denote cyclic shift on $\mathbb{F}_2^n$ and put $D = I + S$.

The linear tail is classical. Periodic additive cellular automata and their matrix powers are treated in [2, 3]; in particular, the Rule-102 power-of-two vanishing mechanism is not claimed here. Repeated-root cyclic-code weight distributions are also available in substantially greater generality [4]. Equality-pattern cellular automata occur in other carriers [1]. We therefore give no contribution credit to D , its kernels and images, homogeneous code enumerators, finite Fourier inversion, or the chromatic polynomial of a cycle. The scoped object is the interface between the literal q -ary rule (1) and that zero-credit tail.

For $w \in \mathcal{A}_q^n$, define its change mask

$$c(w)_i = \mathbf{1}\{w_i \neq w_{i+1}\}.$$

For $0 \leq j \leq n$ and $d \in \mathbb{F}_2^n$, write

$$(2) \quad W_{j,d}(a) = \sum_{\substack{c \in \mathbb{F}_2^n \\ D^j c = d}} a^{\text{wt}(c)},$$

$$(3) \quad C_{n,j}(q) = W_{j,0}(q-1) + (q-1)W_{j,0}(-1).$$

Theorem 1 (finite temporal and inverse atlas). *Let $q \geq 3$ and $n = 2^m \geq 4$.*

(A) *The all-one word is the unique recurrent point. Every nonconstant w has depth*

$$(4) \quad 1 + \min\{j \geq 0 : D^j c(w) = 0\},$$

while the other $q-1$ constant words have depth one. The height is exactly $n+1$. The depth-zero and depth-one shells have sizes 1 and $q-1$; for $1 \leq j \leq n$, the depth- $(j+1)$ shell has size

$$(5) \quad C_{n,j}(q) - C_{n,j-1}(q).$$

At every dyadic $j < n$,

$$(6) \quad C_{n,j}(q) = (1 + (q-1)^{n/j})^j + (q-1)2^j,$$

and the last shell has size

$$(7) \quad \frac{q^n - (q-2)^n}{2} - (q-1)2^{n-1} > 0.$$

(B) *The first image has size $2^n - n$. For $t \geq 2$, put $j = \min(t-1, n)$. Then*

$$(8) \quad \text{im } T_q^t = \mathbf{1} + \text{im } D^j, \quad |\text{im } T_q^t| = 2^{n-j}.$$

For a binary target y , with $d = y + \mathbf{1}$,

$$(9) \quad |(T_q^t)^{-1}(y)| = W_{j,d}(q-1) + (q-1)W_{j,d}(-1), \quad j = \min(t-1, n),$$

and every nonbinary target has fibre zero at positive time.

(C) At time two, a target occurs exactly when $d = y + \mathbf{1}$ has even weight. If $Dc = d$ and $\rho(d) = \min\{\text{wt}(c), n - \text{wt}(c)\}$, then

$$(10) \quad |(T_q^2)^{-1}(y)| = (q-1)^{\rho(d)} + (q-1)^{n-\rho(d)} + 2(q-1)(-1)^{\rho(d)}.$$

For $r < n/2$, precisely $\binom{n}{r}$ supported targets have parameter r ; for $r = n/2$, precisely $\frac{1}{2}\binom{n}{n/2}$ do.

(D) At time $n/2 + 1$, a target occurs exactly when $d = y + \mathbf{1} = (u, u)$. If $h = \text{wt}(u)$, then

$$(11) \quad |(T_q^{n/2+1})^{-1}(\mathbf{1} + d)| = (1 + (q-1)^2)^{n/2-h} (2(q-1))^h + (q-1)2^{n/2}(-1)^h.$$

Exactly $\binom{n/2}{h}$ supported targets have parameter h .

The multiplicities in parts (C) and (D) are parameter-class multiplicities. Equal numerical fibre values from different parameters must be merged. For example, at $n = 4, q = 4$, the $r = 1, 2$ values in (10) coincide.

2. THE NONLINEAR INTERFACE AND TEMPORAL CLOCK

Lemma 2 (change-mask multiplicity). For $c \in \mathbb{F}_2^n$ of weight r , the number of q -ary cyclic words with change mask c is

$$(12) \quad \chi_q(c) = (q-1)^r + (-1)^r(q-1).$$

It vanishes exactly at $r = 1$.

Proof. Contract every equality edge of the cyclic word. If $r > 0$, the remaining quotient is a cycle with r edges, whose proper q -colourings are counted by the right-hand side of (12). If $r = 0$, all letters are equal and both sides equal q . For $q \geq 3$, the asserted zero set follows immediately. $\square$

Lemma 3 (affine Rule-102 tail). For every $t \geq 1$,

$$(13) \quad T_q^t(w) = \mathbf{1} + D^{t-1}c(w).$$

Moreover $D^n = 0$, $\dim \ker D^j = j$, and $\text{im } D^j = \ker D^{n-j}$ for $0 \leq j \leq n$.

Proof. The first identity at $t = 1$ is the definition. For a binary word b , equality is the complement of XOR, so $T_q(b) = \mathbf{1} + Db$. Since $D\mathbf{1} = 0$, induction proves (13). Identifying cyclic words with $\mathbb{F}_2[x]/(x^n - 1)$ identifies D with multiplication by $1 + x$. Dyadicity gives $x^n - 1 = (x + 1)^n$, so multiplication by $x + 1$ is one nilpotent Jordan block. The remaining assertions follow. $\square$

Proof of Theorem 1(A). Lemma 3 gives the pointwise depth formula and absorption by time $n + 1$. Summing Lemma 2 over $\ker D^j$ gives (3); consecutive kernel differences give (5). When j is dyadic, $D^j = I + S^j$ and the kernel consists of j freely chosen bits repeated n/j times. Its weight enumerator is $(1 + a^{n/j})^j$, which yields (6).

Finally $\ker D^{n-1}$ is the even-weight hyperplane, whereas $\ker D^n = \mathbb{F}_2^n$. The q -weighted mass of the complementary odd-weight class is

$$\frac{q^n - (q-2)^n}{2} - (q-1)2^{n-1},$$

where the second term is the signed correction from (12). It is positive for $q \geq 3, n \geq 4$: writing $x = q - 1 \geq 2$ and expanding the odd part gives

$$\frac{(x+1)^n - (x-1)^n}{2} - x2^{n-1} \geq x(nx^{n-2} - 2^{n-1}) \geq x2^{n-2}(n-2) > 0.$$

This proves sharpness. Since every state is absorbed and $\mathbf{1}$ is fixed, it is the sole recurrent point. $\square$

3. IMAGES AND ARBITRARY TARGET FIBRES

Proof of Theorem 1(B). At time one, Lemma 2 says that exactly the complements of the n unit masks are absent, giving $2^n - n$. For $t \geq 2$, every affine class of $\ker D^j$ contains a feasible change mask: the only forbidden masks are units. Here $1 \leq j \leq n$ and $D\mathbf{1} = 0$, hence $\mathbf{1} \in \ker D^j$, including when $j = n$. If a chosen representative is a unit e_i , then $e_i + \mathbf{1}$ has the same D^j -image and weight $n - 1 \geq 3$; Lemma 2 therefore makes it a feasible change mask. Every affine class consequently contains a feasible representative, proving (8).

For a fixed change mask c , Lemma 2 is its exact source multiplicity. Summing it over $D^j c = d$ gives (9). For completeness, character orthogonality also gives the computable expression

$$(14) \quad W_{j,d}(a) = 2^{-n} \sum_{\lambda \in \mathbb{F}_2^n} (-1)^{\lambda \cdot d} (1+a)^{n - \text{wt}((D^j)^T \lambda)} (1-a)^{\text{wt}((D^j)^T \lambda)}.$$

Equation (14) is a verification device, not a separate contribution claim. $\square$

The arbitrary-target formula is exact but does not by itself classify the dependence on d . The next section evaluates two slices rather than hiding that dependence inside a transform.

4. TWO EVALUATED FIBRE SPECTRA

Proof of Theorem 1(C). The image of D is the even-weight hyperplane and its kernel is $\{0, \mathbf{1}\}$. Thus a feasible equation $Dc = d$ has the complementary pair $\{c, c + \mathbf{1}\}$. Substituting their two weights into (12) gives (10); the parity of $\rho(d)$ agrees with both complementary weights because n is even. Complementary pairs of smaller weight $r < n/2$ are indexed by all weight- r masks. At $r = n/2$ each pair is indexed twice. This proves the two class counts. Their sum is 2^{n-1} , and the class-weighted fibre mass is q^n . $\square$

Proof of Theorem 1(D). In characteristic two, $D^{n/2} = (I + S)^{n/2} = I + S^{n/2}$. Hence its image consists exactly of duplicated half-words $d = (u, u)$. For a coordinate pair $(i, i + n/2)$, a zero bit of u requires equal mask bits and contributes $1 + a^2$ to $W_{n/2,d}(a)$; a one bit requires unequal mask bits and contributes $2a$. Therefore

$$W_{n/2,d}(a) = (1 + a^2)^{n/2-h} (2a)^h.$$

Evaluation at $a = q - 1$ and $a = -1$ in (9) gives (11). Choosing the h nonzero coordinates of u gives the class multiplicity, whose sum is $2^{n/2}$; weighted fibre mass again sums to q^n . $\square$

5. EXACT CONTROLS AND LIMITATIONS

The paper-local audit begins with the literal rule (1), not the linearized formulas. It exhausts the six boxes

$$(n, q) = (4, 3), (4, 4), (4, 5), (4, 6), (8, 3), (8, 4),$$

comprising 74,355 words, and checks every orbit, depth shell, image, and binary target fibre. It separately checks both parameter-class spectra for $n \in \{4, 8, 16\}$ and $q \in \{3, 4, 5, 7\}$, including numerical-collision aggregation, mass conservation, and the excluded boundaries $q = 2$, $n = 2$, and nondyadic $n = 6$.

The restrictions matter. At $q = 2$, the feasible change masks differ; at $n = 2$, the support repair by complementary masks fails; at nondyadic length, D need not be nilpotent. Nothing here claims a classification for those regimes, a new theorem about Rule 102, a new repeated-root code enumerator, or absolute novelty. The result is an internally verified, owner-thin finite atlas. External posting, circulation, or submission remains `HOLD_EXTERNAL`.

Data and code statement. No external dataset or training data are used. The exact verifier and its deterministic transcript accompany the paper.

Conflict-of-interest statement. The anonymous author declares no conflict of interest.

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